Project Title	**Mobile Product Segmentation and Recommendation System Using Python and Machine Learning**
Skills take away From This Project	• **Python & Data Processing** • **CSV to DataFrame Conversion** • **Data Cleaning (Handling Missing Values & Duplicates)** • **Data Preprocessing & Feature Engineering** • **Exploratory Data Analysis (EDA)** • **Customer & Product Segmentation using Clustering** • **Feature Scaling & Encoding Techniques** • **Unsupervised Learning (K-Means Clustering)** • **Cluster Analysis & Business Insights Generation** • **Similarity-Based Recommendation System** • **Collaborative Filtering Basics** • **Data Visualization (Seaborn / Matplotlib / Plotly)** • **Web App Development using Streamlit** • **Model Deployment using Streamlit** • **End-to-End Machine Learning Pipeline**
Domain	**E-Commerce Analytics & Intelligent Product Recommendation Systems**

Problem Statement:

The smartphone industry generates large volumes of structured data, including product specifications, pricing, ratings, and user-related information. However, this data is often unorganized and difficult to analyze directly for meaningful insights.

The challenge is to preprocess and transform this data into a structured format and apply clustering techniques to identify distinct groups of products or users. Analyzing these clusters helps in understanding market segments such as premium, mid-range, and budget categories.

Additionally, the project focuses on building a product recommendation system using only structured data to suggest similar mobile products based on their features and user preferences.

This project aims to build a complete **end-to-end data analytics and machine learning pipeline** that enables candidates to:

- Clean and preprocess mobile product data using Python
- Perform exploratory data analysis (EDA) to identify key trends and patterns
- Apply clustering techniques to segment products or users into meaningful groups
- Analyze each cluster to generate actionable business insights
- Build a similarity-based product recommendation system
- Develop an interactive application using Streamlit for visualization and recommendations

Objective:

The objective of this project is to design and implement a **Mobile Product Segmentation and Recommendation System** that follows the complete data lifecycle:

- Data Collection
- Data Cleaning & Preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering & Scaling
- Clustering (Product/User Segmentation)
- Cluster Analysis & Insight Generation
- Recommendation System Development
- Application Development using Streamlit

Approach:

1. Data Collection

- Import the dataset in CSV format
- Load the dataset into Python using Pandas DataFrame
- Understand the structure, columns, and data types

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2. Data Preprocessing

- Handle missing values (null data)
- Remove duplicate records
- Perform data type conversion
- Select relevant features (price, ratings, specifications, engagement score, etc.)

- Encode categorical variables (brand, country, model)
- Standardize and scale the dataset

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3. Exploratory Data Analysis (EDA)

- Analyze product distribution across different brands and countries
- Identify top-rated and low-rated products
- Explore relationships between price, ratings, and specifications
- Identify patterns, trends, and correlations
- Perform statistical summaries and comparisons

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4. Clustering (Segmentation Analysis)

- Apply clustering algorithms (K-Means)
- Assume and create 4 clusters (for example only)
- Group products/users based on similarities
- Analyze each cluster to identify segments (premium, mid-range, budget, etc.)
- Visualize clusters using plots

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5. Recommendation System

- Build a similarity-based recommendation system
- Use product features (price, ratings, specifications)
- Apply similarity techniques (Cosine Similarity / Distance metrics)
- Recommend similar products based on user input or selected product
- Validate recommendation relevance

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6. Application Development & Visualization

- Build an interactive application using Streamlit
- Create visualizations using Plotly / Seaborn / Matplotlib
- Display cluster insights and product segments
- Provide recommendation interface for users

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7. Insights & Reporting

- Generate insights on:
    - Product segmentation (cluster-wise analysis)
    - High-performing and low-performing products
    - Price vs performance trends
    - Customer preference patterns
- Support data-driven decision making

Dataset Link: Global Mobile Reviews Dataset

Results:

- Clean and preprocessed dataset (handled missing values & duplicates)
- Successful CSV to DataFrame conversion
- Well-structured dataset with selected and transformed features
- Effective product/user segmentation using clustering
- Clear identification of segments (premium, mid-range, budget, etc.)
- Cluster-wise analysis with meaningful business insights
- Similarity-based product recommendation system implemented
- Accurate and relevant product recommendations
- Interactive Streamlit application for visualization and recommendations
- Insightful data visualization (Plotly / Seaborn / Matplotlib)
- End-to-end machine learning pipeline implementation

Project Evaluation metrics:

- Proper data preprocessing (handling missing values & duplicates)
- Accurate CSV to DataFrame conversion
- Effective Exploratory Data Analysis (EDA)
- Correct feature selection, encoding, and scaling
- Proper implementation of clustering algorithm (K-Means)
- Appropriate selection of number of clusters
- Clear cluster interpretation and business insights
- Effective visualization of clusters and patterns
- Correct implementation of similarity-based recommendation system

- Relevance and accuracy of recommended products
- Integration of clustering and recommendation logic
- Visualization quality and clarity (Plotly / Seaborn / Matplotlib)
- Streamlit application functionality and user interface
- Clean, modular, and well-structured Python code
- Insight generation and interpretation

Technical Tags:

Python, CSV Data Handling, Data Preprocessing, Exploratory Data Analysis (EDA), Feature Engineering, Data Scaling, Unsupervised Learning, K-Means Clustering, Cluster Analysis, Recommendation Systems, Cosine Similarity, Collaborative Filtering, Data Visualization, Plotly, Seaborn, Matplotlib, Streamlit Application, Machine Learning, E-Commerce Analytics, Product Segmentation.

Project Deliverables:

- CSV dataset (raw data file)
- Cleaned and preprocessed dataset (CSV / DataFrame)
- Python script for data preprocessing and feature engineering
- Implementation of clustering model (K-Means)
- Cluster analysis report with insights (4 segments - for example only)
- Recommendation system implementation (similarity-based)
- Python script for recommendation logic
- Data visualization (Plotly / Seaborn / Matplotlib)
- Interactive Streamlit application (clustering + recommendations)
- EDA report with key findings and patterns
- Final project documentation file

Project Guidelines:

- Follow proper data preprocessing (handle missing values & duplicates)
- Ensure correct CSV to DataFrame conversion
- Perform meaningful Exploratory Data Analysis (EDA)
- Select relevant features and apply proper encoding & scaling
- Implement clustering algorithm correctly (K-Means with 4 - clusters for example only)
- Perform detailed cluster analysis and interpret each segment
- Build an effective similarity-based recommendation system
- Ensure recommendation relevance and logic clarity
- Use appropriate visualization tools (Plotly / Seaborn / Matplotlib)
- Develop a structured Streamlit application (no Power BI)
- Maintain clean, readable, and modular Python code
- Generate clear and actionable insights from analysis
- Provide well-structured documentation

Timeline:

The project must be completed and submitted **within 10 days from the assigned date.**

References:

Project Live Evaluation Guidelines	Project Live Evaluation
Project Excellence Series: Guided Learning & Problem Solving [Python & SQL](Tamil)	Project Excellence Series: Guided L...
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
Streamlit Reference English	[Streamlit Recordings]()
Streamlit Reference Tamil	[Streamlit Recording Tamil]()
Project Orientation (English)	
Project Orientation (Tamil)	Global Mobile Reviews_Tamil.mp4

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.
Note: You can join the doubt session using the link provided in sheet given below

Timing: Monday-Saturday (3:30PM to 4:30PM)

Session Link :[Project Doubt Session [DS/AIML]]

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing: Monday-Saturday (05:30PM to 07:00PM)

Booking link : https://forms.gle/1m2Gsro41fLtZurRA

Created By:	Verified By:	Approved By:
VIGNESH P	Selvamani A	